EXCELLENCE

CLASS 11 · 2026–27

Physics | Lecture - 04

Basic Maths and Calculus

Leftover last class, then ASTC, max–min, phase

By — Excellence Programme

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Topics to be covered

01 Full table 0° to 360° for all six ratios, then arc and radian ↔ degree

02 Last-class homework — slides 21, 23, 25, 33, 35, 37, 49 — answers in simple lines

03 Length prefixes with full names, then compound sin / cos / tan

04 Pending last class, slides 71–82 — small angle, then allied homework

05 ASTC method — sign first, then change the function only at 90° and 270°

06 Maximum and minimum, then phase / phasor

All six ratios from 0° to 90°

θ	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	undef.
cot	undef.	√3	1	1/√3	0
sec	1	2/√3	√2	2	undef.
cosec	undef.	2	√2	2/√3	1

tan 90° and sec 90° are not defined. cosec 0° and cot 0° are not defined.

All six ratios from 90° to 180°

θ	90°	120°	135°	150°	180°
sin	1	√3/2	1/√2	1/2	0
cos	0	−1/2	−1/√2	−√3/2	−1
tan	undef.	−√3	−1	−1/√3	0
cot	0	−1/√3	−1	−√3	undef.
sec	undef.	−2	−√2	−2/√3	−1
cosec	1	2/√3	√2	2	undef.

Second quadrant: only sin and cosec are positive. cot 180° and cosec 180° are not defined.

All six ratios from 180° to 270°

θ	180°	210°	225°	240°	270°
sin	0	−1/2	−1/√2	−√3/2	−1
cos	−1	−√3/2	−1/√2	−1/2	0
tan	0	1/√3	1	√3	undef.
cot	undef.	√3	1	1/√3	0
sec	−1	−2/√3	−√2	−2	undef.
cosec	undef.	−2	−√2	−2/√3	−1

Third quadrant: only tan and cot are positive. tan 270° and sec 270° are not defined.

Rough work / extra work

All six ratios from 270° to 360°

θ	270°	300°	315°	330°	360°
sin	−1	−√3/2	−1/√2	−1/2	0
cos	0	1/2	1/√2	√3/2	1
tan	undef.	−√3	−1	−1/√3	0
cot	0	−1/√3	−1	−√3	undef.
sec	undef.	2	√2	2/√3	1
cosec	−1	−2/√3	−√2	−2	undef.

Fourth quadrant: only cos and sec are positive. 360° repeats 0°. Never write cosec 360°.

Arc length — θ must be in radian

Arc length = R θ

θ in radian

Practical unit of angle: degree. SI unit of angle: radian.
+θ anti-clockwise. −θ clockwise.

If the object turns 90° on a circle of radius R, first change 90° to π/2. Distance = πR/2. Never R × 90°.

The one line: 180° = π rad

Lock this. Every conversion starts here.

180° = π rad

Wherever you see π rad, put 180°.

degree → radian: multiply by π/180
radian → degree: multiply by 180/π

Never write π = 180°. Write π rad = 180°. π with no degree mark is already radian.

Homework slide 21 — six conversions

Last class, slide 21. Convert each. Easy rule: π rad = 180°.

If you see π, put 180°. If you see a degree, multiply by π/180.

π/2 rad = 180°/2 = 90°

π/6 rad = 180°/6 = 30°

90° = 90 × π/180 = π/2 rad

π/4 rad = 180°/4 = 45°

45° = 45 × π/180 = π/4 rad

60° = 60 × π/180 = π/3 rad

Homework slide 23 — radian into degree

Last class, slide 23. Example: 5π/6 means 5 × 180° / 6.

5π/6 = 150°	**2π/3 = 120°**	**4π/3 = 240°**
3π/5 = 108°	**5π = 900°**	**2π/5 = 72°**
7π/12 = 105°	**5π/2 = 450°**	**3π = 540°**

How: cancel. 5π/6 → 5 × 180 / 6 → 5 × 30 = 150. Same for every line.

Rough work / extra work

Homework slide 25 — degree into radian

Last class, slide 25. Multiply the degree by π/180. Cancel 180 if you can.

120° = 2π/3 rad

270° = 3π/2 rad

30° = π/6 rad

135° = 3π/4 rad

150° = 5π/6 rad

π is already radian

Wrong: π° treated as π rad

Right: The board wrote π, not π°. π is already radian. π° would be $\pi^2/180$ rad.

Tiny lengths you will meet in physics

A prefix is a short name stuck in front of metre. It tells you how many metres.

short	full name	in metres	where you meet it
fm	femtometre	10^{-15} m	nucleus of an atom
pm	picometre	10^{-12} m	spacing inside an atom
Å	ångström	10^{-10} m	atom size, X-ray wavelength
nm	nanometre	10^{-9} m	visible light, DNA, thin films
µm	micrometre	10^{-6} m	cell, dust, width of a hair

Å is not SI, but physics still uses it for atoms. Nuclear size is fm, never mm.

Everyday and large lengths

short	full name	in metres	where you meet it
mm	millimetre	10^{-3} m	small lab length, thickness
cm	centimetre	10^{-2} m	ruler, height in class
m	metre	1 m	SI unit of length
km	kilometre	10^{3} m	road, tower, walking distance
Mm	megametre	10^{6} m	Earth radius ≈ 6.4 Mm

1 m = 100 cm **1 km = 1000 m** **1 Å = 10^{-10} m** **1 fm = 10^{-15} m**

How to read a prefix in one line

milli = one thousandth **kilo = one thousand**

micro = one millionth **nano = one billionth**

pico = one trillionth **femto = one thousand-trillionth**

1 mm is 1 m cut into 1000 equal pieces. 1 km is 1000 metres joined.
1 µm is 1 m cut into 1 000 000 pieces. A red-blood-cell is a few µm.

Never mix them: a nucleus is a few fm. An atom is about 1 Å. Light is a few hundred nm.

Compound sine — two angles, one sine

A and B are two angles. Compound means we open the bracket into products.

sin (A + B) = sin A cos B + cos A sin B
In words: sine of a sum = sine-cosine plus cosine-sine.

sin (A − B) = sin A cos B − cos A sin B
Same line, only the middle sign flips. Difference takes a minus.

Do not write sin(A+B) = sin A + sin B. That is the classic trap.

Rough work / extra work

Compound cosine — watch the minus

cos (A + B) = cos A cos B − sin A sin B

In words: cosine of a sum = cosine-cosine minus sine-sine.

cos (A − B) = cos A cos B + sin A sin B

Difference of cosine takes a plus. Sum of cosine takes a minus. Opposite of sine.

Swap those two signs and every 15°, 75°, 105° value will come out wrong.

Compound tangent — one fraction

tan (A + B) = (tan A + tan B) / (1 − tan A tan B)

Top: add the tans. Bottom: 1 minus the product. Sum has a minus downstairs.

tan (A − B) = (tan A − tan B) / (1 + tan A tan B)

Difference: minus on the top, plus downstairs.

If tan A tan B = 1, then tan(A + B) is not defined.

When A and B are the same — double angle

Put B = A in the compound lines. Two of the same angle become 2A.

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

sin 2θ is not 2 sin θ. The 2 sits with the angle, not in front of the sine.

Homework slide 33 — shadow of a pole

Last class, slide 33. Pole height h. Shadow l. Sun at angle θ above the ground. Find l.

1. l = h tan θ
2. l = h / tan θ
3. l = h sin θ
4. l = h / sin θ

Picture: height is the wall, shadow is the floor. tan θ = wall / floor = h / l.

So l = h / tan θ (same as h cot θ)

Option 2 l = h / tan θ

Option 1 swaps height and shadow. tan is not shadow/height.

Homework slide 35 — wire falls

Last class, slide 35. Wire from the top at angle θ breaks and falls. Find x from the base.

1. x = h cot θ
2. x = h tan θ
3. x = h sin θ
4. x = h sec θ

Wire is the long side = h / sin θ. Original base = h / tan θ. After the fall, x is wire minus base.

x = h / sin θ − h / tan θ = h (cosec θ − cot θ)

x = h (cosec θ − cot θ) — none of the four options

Option 1 is only the original base, before the wire falls.

Rough work / extra work

Homework slide 37 — shadow on a slope

Last class, slide 37. Pole 10 m. Slope 30°. Sun 60°. Shadow length l on the slope?

1. 5 m
2. 10 m
3. 10√3 m
4. 10 / (√3 − 1) m

Rise of the slope over length l is l/2. Horizontal run is l√3/2. tan 60° looks at (10 + rise) over run.

√3 = (10 + l/2) / (l √3 / 2) 3l/2 = 10 + l/2 l = 10

Option 2 l = 10 m

Homework slide 49 — four values, easy route

Last class, slide 49. Find sin 120°, cos 150°, sin 240°, cos 120°.

Easy way: write each as 180° with a small angle. ASTC then gives the sign.

(1) sin 120° = sin (180° − 60°) = + sin 60° = √3/2

(2) cos 150° = cos (180° − 30°) = − cos 30° = −√3/2

(3) sin 240° = sin (180° + 60°) = − sin 60° = −√3/2

(4) cos 120° = cos (180° − 60°) = − cos 60° = −1/2

√3/2 **−√3/2** **−√3/2** **−1/2**

Last class stopped at slide 71 — small angle now

Last class ended before small angle. Slides 71 to 82 of that deck are today’s leftover.

small-angle rule → sin 2°, tan 3°, cos 3° → tan 2° trap → allied homework

Then we start the new work: ASTC method, max–min, phase.

When θ is small, the arc looks like the opposite side

On a circle of radius R, a tiny angle makes opposite ≈ arc = Rθ, and adjacent ≈ R.

sin θ ≈ θ **tan θ ≈ θ** **cos θ ≈ 1**

θ in radian. Use only if θ ≤ 5°.

Easy words: change the degree to radian first, then drop the word radian. That number is the answer.

sin 2° is not 2. 2° in radian is 2π/180 = π/90. That is the value.

Pending slide 72 — sin 2°, tan 3°, cos 3°

Last class, slide 72. Using small-angle, find sin 2°, tan 3°, cos 3°.

Step 1: the angle is small, so sine and tan become the radian number. Cosine becomes 1.

(i) $\sin 2^\circ \approx 2 \times (\pi/180) = \pi/90$

(ii) $\tan 3^\circ \approx 3 \times (\pi/180) = \pi/60$

(iii) $\cos 3^\circ \approx 1$

$\sin 2^\circ \approx \pi/90$ $\tan 3^\circ \approx \pi/60$ $\cos 3^\circ \approx 1$

Rough work / extra work

Pending slide 74 — which tan 2° is correct?

Last class, slide 74. Which is correct for tan 2° under small-angle?

1. 2°
2. π / 90
3. π / 30
4. 2 rad

Wrong: tan 2° = 2°

Right: tan 2° ≈ 2 × (π/180) = π/90

Option 2 tan 2° ≈ π/90

Option 1 leaves the degree. Option 4 pretends 2° is already 2 radian. Both are wrong.

Pending slides 76–77 — two allied lines

Last class, slides 76–77. Simplify. (13) $\sin(180^\circ - \theta)$ (14) $\cos(180^\circ + \theta)$

$\sin(180^\circ - \theta) = 0 \cdot \cos\theta - (-1)\sin\theta = \sin\theta$

(13) $\sin(180^\circ - \theta) = \sin\theta$

$\cos(180^\circ + \theta) = (-1)\cos\theta - 0 = -\cos\theta$

(14) $\cos(180^\circ + \theta) = -\cos\theta$

Pending slide 77 — match the column

Last class, slide 77. Match. Modulus means size, ignore the minus.

Column I	easy working	match
(A) sin 390°	same as sin 30° = +1/2	P and T
(B) sin (−30°)	minus, so −1/2	Q and T
(C) cos 120°	minus, so −1/2	Q and T
(D) tan (−120°)	comes out +√3 which is bigger than 1	P and S

1. A → PT, B → QT, C → QT, D → PS
2. A → PT, B → QS, C → QT, D → PS
3. A → QT, B → QS, C → PT, D → PS
4. A → QS, B → PT, C → QT, D → PS

Option 1 A → PT, B → QT, C → QT, D → PS

|√3| > 1 so D is P and S, not T. |1/2| < 1 so A, B, C take T.

Last class, slide 78 — after that class

That class asked you to write the six compound-angle lines ten times.
The two allied lines and the match-column are now answered on the board. Copy them once more.

sin (180° − θ) = sin θ **cos (180° + θ) = − cos θ**

Match: A → PT, B → QT, C → QT, D → PS

Radian and arc — now locked

180° = π rad

Wherever you see π rad, put 180°.

degree → radian: multiply by π/180
radian → degree: multiply by 180/π
Arc length = R θ with θ in radian. Never R × 90°.

90° on a circle of radius R → distance πR/2

Rough work / extra work

Compound and prefixes — now locked

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A + B) = (\tan A + \tan B) / (1 - \tan A \tan B)$

fm femtometre 10^{-15} m **nm nanometre 10^{-9} m** **Å ångström 10^{-10} m**

Do not write sin(A+B) = sin A + sin B. Nuclear size is fm, not mm.

Small angle — the one lock-line

$$\sin\theta \approx \theta \qquad \tan\theta \approx \theta \qquad \cos\theta \approx 1 \qquad (\theta \text{ in radian},\ \theta \le 5^\circ)$$

$\sin 2^\circ \approx \pi/90$ $\tan 3^\circ \approx \pi/60$ $\cos 3^\circ \approx 1$

Never leave the degree number in the approximation. Convert, then drop the word radian.

Leftover from last class is done. After the break: ASTC method, max–min, phase.

BREAK

Leftover from last class is done. Next: ASTC method, max–min, phase

Three steps for any allied angle

θ is acute. The three steps write the value without opening the compound line every time.

1. **Decide the quadrant. Write the sign of that ratio from ASTC.**
2. **At $90° \pm \theta$ and $270° \pm \theta$ the function changes. At $180° \pm \theta$ and $360° \pm \theta$ it stays.**
3. **Change pairs: sin ↔ cos tan ↔ cot sec ↔ cosec**

A S T C → All sin tan cos

Do not look at the sign of the result after the change. Write the ASTC sign first, then the brother.

Where each allied angle sits

I	0° to 90°	All +ve	90° − θ	360° + θ	
II	90° to 180°	sin, cosec +ve	90° + θ	180° − θ	
III	180° to 270°	tan, cot +ve	180° + θ	270° − θ	
IV	270° to 360°	cos, sec +ve	270° + θ	360° − θ	−θ

Rough work / extra work

Allied lines — 90° and 180°

$\sin(90° + \theta) = +\cos\theta$ $\quad$ $\sin(90° - \theta) = +\cos\theta$

$\cos(90° + \theta) = -\sin\theta$ $\quad$ $\cos(90° - \theta) = +\sin\theta$

$\tan(90° - \theta) = +\cot\theta$

$\sin(180° + \theta) = -\sin\theta$ $\quad$ $\sin(180° - \theta) = +\sin\theta$

$\cos(180° + \theta) = -\cos\theta$ $\quad$ $\tan(180° - \theta) = -\tan\theta$

$\cos(90° + \theta)$ is minus sine. That minus is the ASTC sign of cosine in the second quadrant.

Allied lines — 270°, 360°, 450°, 540°

$\sin(270° + \theta) = -\cos\theta$ $\quad$ $\sin(270° - \theta) = -\cos\theta$

$\cos(270° + \theta) = +\sin\theta$ $\quad$ $\tan(360° + \theta) = +\tan\theta$

$\cos(360° - \theta) = +\cos\theta$ $\quad$ $\cos(-\theta) = +\cos\theta$

$\sin(450° + \theta) = \sin(90° + \theta) = +\cos\theta$

$\tan(540° + \theta) = \tan(180° + \theta) = +\tan\theta$

Drop full turns of 360° first. 450° = 90°. 540° = 180°. 720° = 0°.

sin 300° and cos 240°

Find (1) sin 300° and (2) cos 240°. Two routes each.

$\sin 300° = \sin(270° + 30°) = -\cos 30° = -\sqrt{3}/2$

$= \sin(360° - 60°) = -\sin 60° = -\sqrt{3}/2$

(1) $\sin 300° = -\sqrt{3}/2$

$\cos 240° = \cos(180° + 60°) = -\cos 60° = -1/2$

$= \cos(270° - 30°) = -\sin 30° = -1/2$

(2) $\cos 240° = -1/2$

Past 360° — drop full turns first

Find $\sin 405°$, $\cos 480°$ and $\cos 780°$.

$\sin 405° = \sin(360° + 45°) = \sin 45° = 1/\sqrt{2}$

$\cos 480° = \cos(450° + 30°) = \cos(90° + 30°) = -\sin 30° = -1/2$

$\cos 780° = \cos(720° + 60°) = \cos 60° = 1/2$

$\sin 405° = 1/\sqrt{2}$ $\quad$ $\cos 480° = -1/2$ $\quad$ $\cos 780° = 1/2$

720° is two full turns. Cross it out. Do not open a compound line on 780°.

Ten values from ASTC — read down

Find each value. Drop 360° first. Then the three steps.

	angle	reduction	value
(a)	sin 30°	table	1/2
(b)	sin 45°	table	1/√2
(c)	sin 510°	sin (360° + 150°) = sin 150°	1/2
(d)	sin 240°	sin (180° + 60°)	−√3/2
(e)	sin 330°	sin (360° − 30°)	−1/2
(f)	cos 300°	cos (360° − 60°)	1/2
(g)	tan 135°	tan (180° − 45°)	−1
(h)	cot 210°	cot (180° + 30°) = cot 30°	√3
(i)	cos 510°	cos 150° = − cos 30°	−√3/2
(j)	sin 135°	sin (180° − 45°)	1/√2

Rough work / extra work

sin 74° from 2 × 37°

Find sin 74°. Last class: sin 37° = 3/5, cos 37° = 4/5.

sin 74° = sin (2 × 37°) = 2 sin 37° cos 37°

= 2 × (3/5) × (4/5) = 24/25

sin 74° = 24/25

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

sin 22.5° from the half-angle line

Find sin 22.5°. Last class: sin θ = √[(1 − cos 2θ) / 2].

sin 22.5° = √[(1 − cos 45°) / 2] = √[(1 − 1/√2) / 2]
= √[(√2 − 1) / (2√2)]

sin 22.5° = √[(√2 − 1) / (2√2)]

Same line gives cos 15° = (√6 + √2)/4 and cos 105° = (√2 − √6)/4.

BREAK

After the break: maximum and minimum, then phase and phasor

Maximum and minimum of sine and cosine

$(\sin\theta)_{max} = +1$ at $\theta = 90°$

$(\cos\theta)_{max} = +1$ at $\theta = 0°$

$(\sin\theta)_{min} = -1$ at $\theta = 270°$

$(\cos\theta)_{min} = -1$ at $\theta = 180°$

$$-1 \le \sin\theta \le 1$$

$$-1 \le \cos\theta \le 1$$

A constant in front of sine is not the angle. $y = 6 \sin\theta$ has ymax = 6, not $6 \times 90°$.

$y = 6 \sin \theta$

If $y = 6 \sin \theta$, find the maximum and minimum value of y.

$y_{max} = 6 \times (\sin \theta)_{max} = 6 \times 1 = +6$ at $\theta = 90°$

$y_{min} = 6 \times (\sin \theta)_{min} = 6 \times (-1) = -6$ at $\theta = 270°$

$y_{max} = +6$ $y_{min} = -6$

Rough work / extra work

The 3θ is the angle, not the 4

If y = 4 sin (3θ), find ymax and the θ at which y is maximum.

(sin 3θ)max = +1 so ymax = 4 × 1 = +4

The angle that is 90° is 3θ, not θ. 3θ = 90° θ = 30°

ymax = +4 at θ = 30°

Do not set θ = 90°. The sine is of 3θ. Divide 90° by 3.

Range of a projectile

If $R = u^2 \sin(2\theta) / g$, with u and g constant, find θ at which R is maximum.

Rmax when $\sin(2\theta)$ is maximum, so $\sin(2\theta) = 1$ $2\theta = 90°$ $\theta = 45°$

$\theta = 45°$ $R_{max} = u^2 / g$

This is why a level-ground projectile goes farthest at 45°.

Add like sines first

If $y = 3\sin\theta + 4\sin\theta$, find ymax.

$y = 7\sin\theta$ $\quad$ $y_{max} = 7 \times 1 = 7$ $\quad$ at $\theta = 90°$

ymax = 7

Do not open $\sqrt{(3^2+4^2)}$ here. Both terms are sine of the same angle. Add the 3 and the 4.

A constant sitting next to the sine

Find ymax and ymin.

$y = 3 + 2\sin\theta$ $\quad y_{max} = 3 + 2(1) = 5$ $\quad y_{min} = 3 + 2(-1) = 1$

$y = 3 + 4\sin\theta$ $\quad y_{max} = 7$ $\quad y_{min} = 3 - 4 = -1$

$y = 3\sin\theta - 4$ $\quad y_{max} = 3 - 4 = -1$ $\quad y_{min} = -3 - 4 = -7$

$3 + 2\sin\theta \rightarrow 5$ and 1 $\quad 3 + 4\sin\theta \rightarrow 7$ and -1 $\quad 3\sin\theta - 4 \rightarrow -1$ and -7

When sine and cosine are added

sin θ and cos θ do not peak together. The combined amplitude is the hypotenuse of a and b.

$y = a \sin\theta + b \cos\theta$

$$y_{max} = +\sqrt{(a^2 + b^2)} \qquad y_{min} = -\sqrt{(a^2 + b^2)}$$

Do not write ymax = a + b. That would need sin = 1 and cos = 1 at the same instant, which never happens.

Rough work / extra work

$y = 3 \sin \theta + 4 \cos \theta$

If $y = 3 \sin \theta + 4 \cos \theta$, find ymax and ymin.

$y_{max} = \sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = +5$

$y_{min} = -5$

$y_{max} = +5 \qquad y_{min} = -5$

Minimum magnitude of a fraction

If $y = 4 / (3 \sin \theta + 2 \cos \theta)$, find the minimum magnitude of y.

4/10 = 0.4. 4/100 = 0.04. A larger denominator makes a smaller positive fraction.

$y_{min} = 4 / (3 \sin \theta + 2 \cos \theta)_{max} = 4 / \sqrt{(3^2 + 2^2)} = 4 / \sqrt{13}$

minimum magnitude $= 4 / \sqrt{13}$

Five max–min — every line

Find maximum and minimum of each.

	y	ymax	ymin
(a)	$4\sin\theta + 3\cos\theta$	$\sqrt{(16+9)} = 5$	-5
(b)	$2\sin\theta - 3\cos\theta$	$\sqrt{(4+9)} = \sqrt{13}$	$-\sqrt{13}$
(c)	$1 + 5\sin\theta$	$1+5 = 6$	$1-5 = -4$
(d)	$1 - 2\sin\theta$	$1-2(-1) = 3$	$1-2(1) = -1$
(e)	$5\sin\theta - 12\cos\theta$	$\sqrt{(25+144)} = 13$	-13

Phase is the angle inside the ratio

Phase is the angle sitting inside the trigonometric function. Not the 4 in front.

$\sin \theta$	$\text{phase} = \theta$
$\sin (\theta + \alpha)$	$\text{phase} = \theta + \alpha$
$\sin (\omega t + \alpha + 90°)$	$\text{phase} = \omega t + \alpha + 90°$

Later: AC, SHM, waves, wave optics, EM waves, vectors.

Phase difference — same function first

Angle difference between two of the same trigonometric function is called phase difference.

$y_1 = \sin\theta$ **$\varphi_1 = \theta$**

$y_2 = \sin(\theta + 60°)$ **$\varphi_2 = \theta + 60°$**

$\Delta\varphi = \varphi_2 - \varphi_1 = +60°$ **(y2 leads y1)** **$\Delta\varphi = \varphi_1 - \varphi_2 = -60°$** **(lag)**

Do not subtract a cosine phase from a sine phase until both are written as the same ratio.

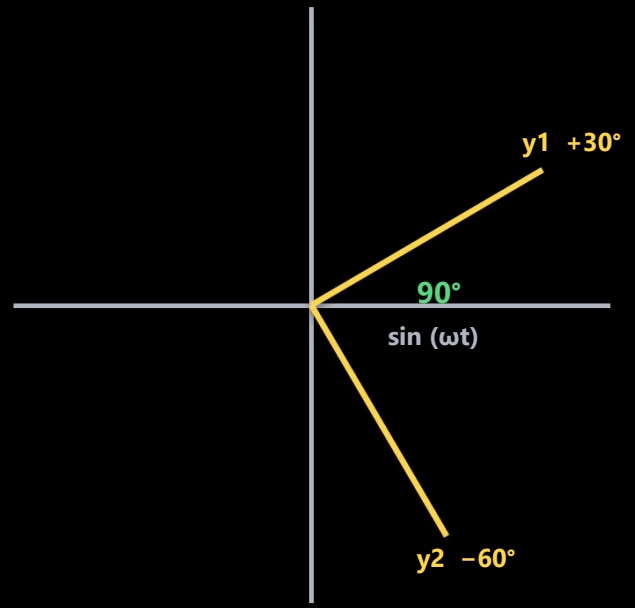
Rough work / extra work

y1 leads y2 by 90°

Find the phase difference. $y1 = \sin(\omega t + 30°)$ $y2 = \sin(\omega t - 60°)$

$\Delta\varphi = \varphi1 - \varphi2 = (\omega t + 30°) - (\omega t - 60°) = 90°$

$\Delta\varphi = 90°$ — y1 is ahead of y2



Convert to the same ratio, or the answer is wrong

Find Δφ. y1 = 3 cos (ωt − 90°) y2 = 4 sin (ωt)

Wrong: Δφ = ωt − (ωt − 90°) = 90°

Right: first write both as sine (or both as cosine)

cos (ωt − 90°) = sin ωt so y1 = 3 sin (ωt)

Δφ = 0 — they are in phase

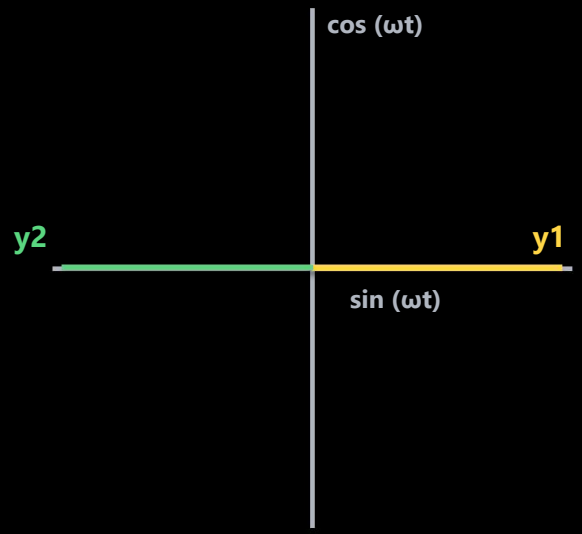
A minus cosine is a sine with a shift. Convert, then subtract.

cos (ωt + π/2) points opposite to sine

Find Δφ. y1 = 3 sin (ωt) y2 = 4 cos (ωt + π/2)

cos (ωt + π/2) = − sin (ωt) so y2 = − 4 sin (ωt)

Δφ = 180° = π rad — opposite on the phasor



Two more phase differences

(1) $x_1 = A\cos(\omega t - \pi/2)$ $y_2 = A\cos(\omega t + \pi/3)$

Same function already. $\Delta\varphi = (\omega t + \pi/3) - (\omega t - \pi/2) = 5\pi/6 = 150°$

(1) $\Delta\varphi = 150°$

(2) $y_1 = 4\sin(\theta + \pi/4)$ $y_2 = 3\sin(\theta - \pi/2)$

$\Delta\varphi = (\theta + \pi/4) - (\theta - \pi/2) = \pi/4 + \pi/2 = 135°$

(2) $\Delta\varphi = 135°$ Also: $y_1 = 4\sin(\theta+\pi/3)$, $y_2 = 3\sin(\theta+\pi/6) \rightarrow 30°$

AC current and voltage

$I = I_0 \sin(\omega t - 30°)$ $V = V_0 \cos(\omega t)$. Find the phase difference.

$V = V_0 \cos(\omega t) = V_0 \sin(\omega t + 90°)$

$\Delta\varphi = 90° - (-30°) = 120° = 2\pi/3$

$\Delta\varphi = 2\pi/3$ — voltage leads current by 120°

Rough work / extra work

Negative cosine is a shifted sine

$I = -I_0 \cos(\omega t)$ $V = V_0 \sin(\omega t - \pi/6)$. Find $\Delta\varphi$.

1. $2\pi/3$
2. $\pi/3$
3. $\pi/6$
4. $5\pi/6$

$-\cos(\omega t) = \sin(\omega t - \pi/2)$ $\Delta\varphi = (-\pi/6) - (-\pi/2) = \pi/3$

Option 2 $\Delta\varphi = \pi/3$

Person walks toward a 40 m building

A person sees a 40 m building at elevation 30°, then at 60°. Distance moved toward the building?

1. $40/\sqrt{3}$
2. $80/\sqrt{3}$
3. $60/\sqrt{3}$
4. $120/\sqrt{3}$

$x1 = 40 \cot 30° = 40\sqrt{3}$ $\quad x2 = 40 \cot 60° = 40/\sqrt{3}$

$d = x1 - x2 = 40\sqrt{3} - 40/\sqrt{3} = 40(3 - 1)/\sqrt{3} = 80/\sqrt{3}$

Option 2 $\quad d = 80/\sqrt{3}$

$2\,(\sin 15° + \sin 75°)^2$

What is the value of $2\,(\sin 15° + \sin 75°)^2$?

$\sin 75° = \cos 15°$ so $2\,(\sin 15° + \cos 15°)^2$

$= 2\,(\sin^2 15° + \cos^2 15° + 2 \sin 15° \cos 15°) = 2\,(1 + \sin 30°)$

$= 2\,(1 + 1/2) = 3$

$2\,(\sin 15° + \sin 75°)^2 = 3$

If sin θ = 1/2, first quadrant

If sin θ = 1/2, the value of √3 (cos θ + 2 tan θ) is:

1. 7
2. 15/2
3. 7/2
4. none

First quadrant: θ = 30° cos 30° = √3/2 tan 30° = 1/√3

√3 (√3/2 + 2/√3) = 3/2 + 2 = 7/2

Option 3 7/2

If $\sin\theta = \sqrt{5}/3$

If $\sin\theta = \sqrt{5}/3$, find $\tan\theta + 4\sec\theta$. Take θ acute.

$\cos\theta = \sqrt{(1 - 5/9)} = \sqrt{(4/9)} = 2/3$ $\tan\theta = (\sqrt{5}/3) / (2/3) = \sqrt{5}/2$

$\sec\theta = 3/2$ $\tan\theta + 4\sec\theta = \sqrt{5}/2 + 4 \times (3/2) = \sqrt{5}/2 + 6$

$\tan\theta + 4\sec\theta = 6 + \sqrt{5}/2$

Rough work / extra work

$\tan^{-1}(1)$ — principal value

The value of $\tan^{-1}(1)$ is

1. $\pi/4$
2. $5\pi/4$
3. both (1) and (2)
4. none of these

Wrong: $\tan(5\pi/4) = 1$, so both

Right: $\tan^{-1}$ returns only the principal value, in $(-\pi/2, \pi/2)$

Option 1 $\pi/4$

$5\pi/4$ is an angle whose tangent is 1. It is not the inverse-tangent of 1.

Phase difference table — copy, do not wait

Find the phase difference in each row. Convert to the same trigonometric function first.

Equation 1	Equation 2	Δφ
I0 sin(θ + π/3)	I0 sin(θ − π/6)	
I0 sin(θ + π/3)	I0 cos(θ + π/3)	
4 sin(ωt)	4 cos(ωt − π/3)	
3 sin(ωt + π/2)	5 cos(ωt)	
4 cos(ωt − π/2)	4 sin(ωt + π/2)	
3 sin(ωt + π/3)	3 cos(ωt)	
4 sin(θ − 60°)	4 cos(θ − 30°)	

After class

Daily test before the next lecture.

Write the three ASTC steps ten times on a rough page.

Write $y_{max} = \sqrt{(a^2 + b^2)}$ for $y = a \sin \theta + b \cos \theta$ ten times.

Tonight: the seven phase-difference rows. Convert first, then subtract.

ASTC method — now locked

1. Sign from ASTC. 2. Function changes at 90° and 270° only. 3. sin ↔ cos.

$\sin 300° = -\sqrt{3}/2$ $\cos 480° = -1/2$ $\sin 405° = 1/\sqrt{2}$

$\sin 74° = 24/25$ $\sin 22.5° = \sqrt{[(\sqrt{2} - 1)/(2\sqrt{2})]}$

Drop full turns of 360° first.

Max–min — the two boxes

$y = A \sin(n\theta)$ $\quad y_{max} = |A|$ $\quad$ at $n\theta = 90°$

$$y = a \sin\theta + b \cos\theta \qquad y_{max} = \sqrt{(a^2 + b^2)}$$

$3 \sin\theta + 4 \cos\theta \rightarrow \pm 5$ $\quad \frac{4}{(3 \sin\theta + 2 \cos\theta)} \rightarrow \frac{4}{\sqrt{13}}$

3 sin θ + 4 sin θ is 7 sin θ, not √(9+16). Add like ratios first.

Rough work / extra work

Phase — the one lock-line

Phase is the angle inside the ratio. Phase difference needs the same trigonometric function.

$\Delta\varphi = \varphi_2 - \varphi_1$ **after both are sine, or both are cosine**

$3\cos(\omega t - 90°)$ and $4\sin(\omega t)$ → $\Delta\varphi = 0$

Never subtract a cosine-phase from a sine-phase. Convert, then subtract.

Daily test before the next lecture. Seven phase rows tonight.